STOCHASTIC FINANCE AND RISK MODELLING
TD 1 & 2 - Step-by-Step Solutions

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1 Exercise 1: Conditional Expectation

Problem Statement: Let X be a random variable taking values in some finite set and $Y \in L^1(\Omega, \mathcal{F}, P)$. Prove that there exist $n \in \mathbb{N}$ and a partition $(A_1, \ldots, A_n)$ of Ω such that:

$$E[Y|X] = \sum_{i=1}^{n} \frac{E[Y 1_{A_i}]}{P[A_i]} 1_{A_i} \quad P\text{-a.s.}$$

Step 1: Constructing the Partition

We are told that X takes values in a finite set. Let's call these distinct values $\{a_1, a_2, \ldots, a_n\}$. By definition, this means that for any $i \neq j$, $a_i \neq a_j$.

We can group the outcomes in our sample space Ω based on the value X takes. Let's define the set A_i as the event where X equals a_i:

$$A_i = X^{-1}(a_i) = \{\omega \in \Omega \mid X(\omega) = a_i\}$$

for $i \in \{1, \ldots, n\}$.

Because X must take exactly one of these values for any given outcome ω, the sets $A_1, \ldots, A_n$ do not overlap (they are mutually exclusive) and together they cover the entire sample space Ω. Therefore, $(A_1, \ldots, A_n)$ forms a valid partition of Ω.

We can also write X using these partition sets:

$$X = \sum_{i=1}^{n} a_i 1_{A_i}$$

Step 2: Checking the Measurability Condition

To prove that our proposed formula is indeed the conditional expectation $E[Y|X]$, it must satisfy two mathematical conditions. The first is that the proposed random variable must be $\sigma(X)$-measurable.

The σ-algebra generated by X, denoted $\sigma(X)$, is exactly the σ-algebra generated by our partition: $\sigma(X) = \sigma(A_1, \ldots, A_n)$.

Let's look at our proposed formula:

$$Z = \sum_{i=1}^{n} \frac{E[Y 1_{A_i}]}{P[A_i]} 1_{A_i}$$

Notice that $\frac{E[Y 1_{A_i}]}{P[A_i]}$ is simply a constant (a real number) for each i. Since the indicator functions 1_{A_i} are $\sigma(X)$-measurable (because $A_i \in \sigma(X)$), any linear combination of them is also $\sigma(X)$-measurable. Thus, the first condition is satisfied.

Step 3: Checking the Partial Expectation Condition

The second condition for conditional expectation requires that for any set $A \in \sigma(X)$, the expectation of our proposed formula over the set A equals the expectation of Y over the set A. Mathematically, for all $A \in \sigma(X)$:

$$E[Z 1_A] = E[Y 1_A]$$

Since $\sigma(X)$ is generated by the partition $(A_1, \ldots, A_n)$, it is sufficient to check this condition just for the building blocks of the σ-algebra: the partition sets A_j for $j \in \{1, \ldots, n\}$.

Let's plug our formula Z into the left side of the equation:

$$E[Z 1_{A_j}] = E\left[\left(\sum_{i=1}^{n} \frac{E[Y 1_{A_i}]}{P[A_i]} 1_{A_i}\right) 1_{A_j}\right]$$

By the linearity of the expectation operator, we can move the expectation inside the sum:

$$= \sum_{i=1}^{n} \frac{E[Y1_{A_i}]}{P[A_i]} E[1_{A_i} 1_{A_j}]$$

Now, remember that our sets A_i and A_j are disjoint if $i \neq j$. Therefore, the product of their indicator functions $1_{A_i} 1_{A_j} = 0$ everywhere except when $i = j$. When $i = j$, $1_{A_j} 1_{A_j} = 1_{A_j}$. This makes the entire sum collapse into a single term where $i = j$:

$$= \frac{E[Y1_{A_j}]}{P[A_j]} E[1_{A_j}]$$

Since $E[1_{A_j}]$ is simply the probability of the event A_j, we have $E[1_{A_j}] = P[A_j]$. Substituting this in gives:

$$= \frac{E[Y1_{A_j}]}{P[A_j]} P[A_j] = E[Y1_{A_j}]$$

We have successfully shown that $E[Z1_{A_j}] = E[Y1_{A_j}]$, which satisfies the second condition. Therefore, our formula Z is indeed $E[Y|X]$.

2 Exercise 2: Martingales and Casino Strategies

Context: A player plays a sequence of independent games. Let $X_1, X_2, \ldots$ be the algebraic gain (gain minus initial bet) at each game. We assume $\mathbb{E}[|X_i|] < \infty$, the game is fair ($\mathbb{E}[X_i] = 0$), and $S_n = X_1 + \cdots + X_n$ is the total winnings – or losses – after n games. The filtration is given by $\mathcal{F}_n = \sigma(S_1, \ldots, S_n) = \sigma(X_1, \ldots, X_n)$.

2.1 Part 1: The Standard Game

Question 1: What does $\mathcal{F}_n$ represent?

Step-by-Step Explanation: The σ-algebra $\mathcal{F}_n$ represents the total accumulation of information available to the player exactly after the n-th game has been played. If a random variable is $\mathcal{F}_n$-measurable, it means its value is completely known by time n. In this casino context, it perfectly represents the history of all the gains & losses up to game n.

Question 2: Compute $\mathbb{E}[S_{n+1}|\mathcal{F}_n]$.

Step-by-Step Explanation: We want to find the expected total winnings after game $n+1$, given all the information up to game n.

First, let's write S_{n+1} in terms of S_n:

$$S_{n+1} = S_n + X_{n+1}$$

Now, we apply the conditional expectation:

$$\mathbb{E}[S_{n+1}|\mathcal{F}_n] = \mathbb{E}[S_n + X_{n+1}|\mathcal{F}_n]$$

By the linearity property of conditional expectations, we can split the sum:

$$= \mathbb{E}[S_n|\mathcal{F}_n] + \mathbb{E}[X_{n+1}|\mathcal{F}_n]$$

Now we evaluate each term based on what we know at time n: 1. Since S_n is completely determined by the information at time n, it is $\mathcal{F}_n$-measurable. We can just pull it out of the expectation: $\mathbb{E}[S_n|\mathcal{F}_n] = S_n$. 2. The games are independent, meaning the outcome of game $n+1$ (X_{n+1}) does not depend on the history of the games up to time n ($\mathcal{F}_n$). Therefore, conditioning on $\mathcal{F}_n$ gives no new information about X_{n+1}: $\mathbb{E}[X_{n+1}|\mathcal{F}_n] = \mathbb{E}[X_{n+1}]$.

Substitute these back into our equation:

$$= S_n + \mathbb{E}[X_{n+1}]$$

Finally, because the game is fair, we know $\mathbb{E}[X_{n+1}] = 0$.

$$= S_n + 0 = S_n \quad P\text{-a.s.}$$

This proves that the total winnings process S_n is a martingale.

2.2 Part 2: Introducing a Betting Strategy

Context: The player now chooses a bet amount H_i at the i-th game. The total winnings become $\tilde{S}_n = \sum_{i=1}^{n} H_i X_i$.

Question 1: What is the reasonable assumption about the measurability and integrability of H_i?

Step-by-Step Explanation: In reality, a player must decide their bet for game i *before* the game is played. They cannot use a crystal ball to look into the future. Therefore, the bet H_i can only depend on the information available strictly before game i, which is $\mathcal{F}_{i-1}$.

Mathematically, we say H_i must be a *predictable* process. For integrability, we typically assume H_i is bounded so that the resulting stochastic integral remains integrable.

Question 2: Compute $\mathbb{E}[\tilde{S}_{n+1}|\mathcal{F}_n]$ and comment.

Step-by-Step Explanation: Just like before, we write the wealth at $n+1$ as the wealth at n plus the change at $n+1$:

$$\tilde{S}_{n+1} = \tilde{S}_n + H_{n+1}X_{n+1}$$

Apply the conditional expectation:

$$\mathbb{E}[\tilde{S}_{n+1}|\mathcal{F}_n] = \mathbb{E}[\tilde{S}_n|\mathcal{F}_n] + \mathbb{E}[H_{n+1}X_{n+1}|\mathcal{F}_n]$$

Again, $\tilde{S}_n$ is perfectly known at time n, so $\mathbb{E}[\tilde{S}_n|\mathcal{F}_n] = \tilde{S}_n$.

For the second term, we use the property of "pulling out what is known". Since H_{n+1} is predictable, it is decided based on $\mathcal{F}_n$. Thus, it acts as a constant relative to $\mathcal{F}_n$ and can be pulled out of the conditional expectation:

$$\mathbb{E}[H_{n+1}X_{n+1}|\mathcal{F}_n] = H_{n+1}\mathbb{E}[X_{n+1}|\mathcal{F}_n]$$

Because X_{n+1} is independent of $\mathcal{F}_n$, we drop the condition:

$$= H_{n+1}\mathbb{E}[X_{n+1}]$$

Since the game is fair ($\mathbb{E}[X_{n+1}] = 0$):

$$= H_{n+1} \times 0 = 0$$

Putting it all together:

$$\mathbb{E}[\tilde{S}_{n+1}|\mathcal{F}_n] = \tilde{S}_n + 0 = \tilde{S}_n \quad P\text{-a.s.}$$

Comment: This proves that you cannot turn a fair game into a winning game simply by varying your bet sizes based on past information. The process remains a martingale.

2.3 Part 3: Stopping Times

Context: The player stops at a bounded random time $\tau : \Omega \to \mathbb{N}$.

Question 1: What is the reasonable assumption about the measurability of τ?

Step-by-Step Explanation: The decision to stop playing right after game n (the event $\{\tau = n\}$) must be made using only the information available up to and including game n. You cannot stop today because you know you will lose tomorrow. Therefore, the event $\{\tau = n\}$ must belong to the σ-algebra $\mathcal{F}_n$. This is the formal definition of a stopping time.

Question 2: Assume $\{\tau = i\}$ is independent of $\mathcal{F}_i$ for $i \geq 1$. Compute $\mathbb{E}[S_\tau]$.

Step-by-Step Explanation: Since τ is bounded, there exists some maximum integer k such that $\tau \leq k$ always happens ($P(\tau \leq k) = 1$). We can rewrite the value of the stopped process S_τ by summing over all possible stopping times using indicator functions:

$$S_\tau = \sum_{n=1}^{k} \mathbb{I}_{\{\tau=n\}} S_n$$

Now take the expected value of both sides:

$$\mathbb{E}[S_\tau] = \mathbb{E}\left[\sum_{n=1}^{k} \mathbb{I}_{\{\tau=n\}} S_n\right]$$

By linearity (since the sum is finite), we can pull the sum outside the expectation:

$$= \sum_{n=1}^{k} \mathbb{E}[\mathbb{I}_{\{\tau=n\}} S_n]$$

We are given that $\{\tau = n\}$ is independent of $\mathcal{F}_n$. Since S_n is $\mathcal{F}_n$-measurable, the indicator function $\mathbb{I}_{\{\tau=n\}}$ and the random variable S_n are independent. For independent variables, the expectation of their product is the product of their expectations:

$$= \sum_{n=1}^{k} \mathbb{E}[\mathbb{I}_{\{\tau=n\}}]\mathbb{E}[S_n]$$

We know that $\mathbb{E}[\mathbb{I}_{\{\tau=n\}}] = P(\tau = n)$. We also know from Part 1 that $\mathbb{E}[S_n] = \mathbb{E}[X_1 + \cdots + X_n] = 0$ because each game is fair.

$$= \sum_{n=1}^{k} P(\tau = n) \times 0 = 0$$

Thus, $\mathbb{E}[S_\tau] = 0$.

3 Exercise 3: Submartingales and Convex Functions

Problem Statement: Assume that $M = (M_t)_{t\in\mathbb{N}}$ is a positive $\mathbb{F}$-submartingale such that $M_t \in L^2$ for all $t \geq 0$. Let $N_t = M_t^2$. Show that N is an $\mathbb{F}$-submartingale.

Step 1: Check Measurability and Integrability

To prove that N_t is a submartingale, we first need to ensure it satisfies the basic properties of a stochastic process: it must be adapted to the filtration and integrable.

1. **Measurability (Adaptedness):** We are given that M_t is $\mathcal{F}_t$-measurable. Because the function $f(x) = x^2$ is a continuous function, applying it to a measurable random variable results in another measurable random variable. Therefore, $N_t = M_t^2$ is $\mathcal{F}_t$-measurable.

2. **Integrability:** We are given that $M_t \in L^2$, which by definition means that the expected value of its square is finite: $\mathbb{E}[M_t^2] < \infty$. Since $N_t = M_t^2$, this immediately gives us $\mathbb{E}[|N_t|] = \mathbb{E}[M_t^2] < \infty$, so N_t is integrable.

Step 2: Prove the Submartingale Property

We need to show that for $t \geq s$, we have $\mathbb{E}[N_t|\mathcal{F}_s] \geq N_s$. There are two distinct ways to prove this, both hinted at in your class correction.

Method 1: Using Jensen's Inequality (The elegant way)

Jensen's Inequality states that for any convex function ϕ and a random variable X, $\mathbb{E}[\phi(X)|\mathcal{F}] \geq \phi(\mathbb{E}[X|\mathcal{F}])$.

Let $\phi(x) = x^2$, which is strictly convex. Applying Jensen's inequality to our process:

$$\mathbb{E}[M_t^2|\mathcal{F}_s] \geq (\mathbb{E}[M_t|\mathcal{F}_s])^2$$

Now, we use the fact that M is a positive submartingale. The submartingale property tells us:

$$\mathbb{E}[M_t|\mathcal{F}_s] \geq M_s$$

Because M_s is a positive process ($M_s \geq 0$), squaring both sides preserves the inequality (the function x^2 is increasing for $x \geq 0$):

$$(\mathbb{E}[M_t|\mathcal{F}_s])^2 \geq M_s^2$$

Combining our two inequalities:

$$\mathbb{E}[N_t|\mathcal{F}_s] = \mathbb{E}[M_t^2|\mathcal{F}_s] \geq M_s^2 = N_s$$

This concludes the proof using Method 1.

Method 2: Algebraic Expansion (The computational way)
Let's look at the expected value of the squared difference $(M_t - M_s)^2$ conditioned on $\mathcal{F}_s$. Since a square is always non-negative, its expected value must be non-negative:

$$\mathbb{E}[(M_t - M_s)^2|\mathcal{F}_s] \geq 0$$

Now, let's expand the square inside the expectation:

$$\mathbb{E}[M_t^2 - 2M_tM_s + M_s^2|\mathcal{F}_s] \geq 0$$

Using the linearity of conditional expectation, we split the terms:

$$\mathbb{E}[M_t^2|\mathcal{F}_s] - \mathbb{E}[2M_tM_s|\mathcal{F}_s] + \mathbb{E}[M_s^2|\mathcal{F}_s] \geq 0$$

Now we evaluate what is "known" at time s. Since M_s is $\mathcal{F}_s$-measurable, it acts as a constant under the conditioning on $\mathcal{F}_s$ and can be pulled out:

$$\mathbb{E}[M_t^2|\mathcal{F}_s] - 2M_s\mathbb{E}[M_t|\mathcal{F}_s] + M_s^2 \geq 0$$

Next, we isolate $\mathbb{E}[M_t^2|\mathcal{F}_s]$, which is $\mathbb{E}[N_t|\mathcal{F}_s]$:

$$\mathbb{E}[M_t^2|\mathcal{F}_s] \geq 2M_s\mathbb{E}[M_t|\mathcal{F}_s] - M_s^2$$

Because M is a submartingale, we know $\mathbb{E}[M_t|\mathcal{F}_s] \geq M_s$. Since $M_s \geq 0$, multiplying by $2M_s$ preserves the inequality:

$$2M_s\mathbb{E}[M_t|\mathcal{F}_s] \geq 2M_s(M_s) = 2M_s^2$$

Substitute this back into our isolated equation:

$$\mathbb{E}[M_t^2|\mathcal{F}_s] \geq 2M_s^2 - M_s^2$$

$$\mathbb{E}[M_t^2|\mathcal{F}_s] \geq M_s^2$$

Substituting N back in, we get $\mathbb{E}[N_t|\mathcal{F}_s] \geq N_s$, which proves N is a submartingale.

4 Exercise 4: Doob's Decomposition

Context: Consider an $\mathbb{F}$-adapted process $S = (S_n)_{n\in\mathbb{N}}$.

4.1 Part 1: The General Decomposition

Assume that the process S is integrable. We define a new process A by setting $A_0 := 0$ and for each $n \geq 1$:

$$A_n := \sum_{k=1}^{n} E[S_k - S_{k-1}|\mathcal{F}_{k-1}]$$

Question 1(a): Show that $(A_n)_{n\in\mathbb{N}}$ is $\mathbb{F}$-predictable.

Step-by-Step Explanation: A process is predictable if its value at time n is completely determined by the information available at time $n-1$. Let's look at the terms in the sum that defines A_n. The index k runs from 1 up to n. The very last term in this sum is $E[S_n - S_{n-1}|\mathcal{F}_{n-1}]$. By definition, taking the conditional expectation given $\mathcal{F}_{n-1}$ results in a random variable that is $\mathcal{F}_{n-1}$-measurable. Since all previous terms (for $k < n$) are conditioned on even earlier information, they are also $\mathcal{F}_{n-1}$-measurable. Because A_n is a sum of $\mathcal{F}_{n-1}$-measurable terms, A_n itself is $\mathcal{F}_{n-1}$-measurable. Therefore, A is predictable.

Question 1(b): Show that $S_n = S_0 + A_n + M_n$ for all $n \in \mathbb{N}$, where M is an $\mathbb{F}$-martingale.

Step-by-Step Explanation: We are forced to define M_n by rearranging the target equation:

$$M_n = S_n - S_0 - A_n$$

We need to prove this M_n is a martingale. This means checking $E[M_n|\mathcal{F}_{n-1}] = M_{n-1}$. Let's calculate $E[M_n|\mathcal{F}_{n-1}]$:

$$E[M_n|\mathcal{F}_{n-1}] = E[S_n - S_0 - A_n|\mathcal{F}_{n-1}]$$

By linearity, and knowing S_0 and A_n are known at time $n-1$:

$$= E[S_n|\mathcal{F}_{n-1}] - S_0 - A_n$$

Now, let's rewrite A_n by pulling out its last term:

$$A_n = A_{n-1} + E[S_n - S_{n-1}|\mathcal{F}_{n-1}] = A_{n-1} + E[S_n|\mathcal{F}_{n-1}] - S_{n-1}$$

Substitute this expression for A_n back into our martingale equation:

$$E[M_n|\mathcal{F}_{n-1}] = E[S_n|\mathcal{F}_{n-1}] - S_0 - (A_{n-1} + E[S_n|\mathcal{F}_{n-1}] - S_{n-1})$$

The $E[S_n|\mathcal{F}_{n-1}]$ terms cancel out perfectly:

$$= S_{n-1} - S_0 - A_{n-1}$$

Looking at our definition of M, this is exactly M_{n-1}. Thus, M is a martingale.

Question 1(c): Show that the above decomposition into A and M is (P-a.s.) unique.

Step-by-Step Explanation: Suppose we have two such decompositions for the same process S: $S_n = S_0 + A_n + M_n$ $S_n = S_0 + A'_n + M'_n$ Subtract the second equation from the first: $0 = (A_n - A'_n) + (M_n - M'_n)$ Rearranging gives: $M_n - M'_n = A'_n - A_n$ The left side is the difference of two martingales, which is still a martingale. The right side is the difference of two predictable processes, which is still predictable. This means the process $Y_n = M_n - M'_n$ is both a martingale & predictable. Because it is a martingale, $E[Y_n|\mathcal{F}_{n-1}] = Y_{n-1}$. Because it is predictable, $E[Y_n|\mathcal{F}_{n-1}] = Y_n$. Therefore, $Y_n = Y_{n-1}$. The process is constant over time. Since both M and M' start at 0, their difference is 0 at time 0. Thus, $Y_n = 0$ for all n. This means $M_n = M'_n$ and $A_n = A'_n$, proving uniqueness.

Question 1(d): Summarize the result. Every integrable adapted process S can be uniquely written as the sum of its initial value, a predictable process starting at zero, and a martingale starting at zero.

4.2 Part 2: Submartingale Case

Assume further that S is an $\mathbb{F}$-submartingale.

Question 2(a): Show that $(A_n)_{n\in\mathbb{N}}$ is non-decreasing.

Step-by-Step Explanation: Let's look at the difference between consecutive terms of A: $A_n - A_{n-1} = E[S_n - S_{n-1}|\mathcal{F}_{n-1}] = E[S_n|\mathcal{F}_{n-1}] - S_{n-1}$ Because S is a submartingale, we know by definition that

$E[S_n|\mathcal{F}_{n-1}] \geq S_{n-1}$. Therefore, $E[S_n|\mathcal{F}_{n-1}] - S_{n-1} \geq 0$. Since $A_n - A_{n-1} \geq 0$, the process A is non-decreasing.

Question 2(b): Write down Doob's decomposition of $\mathbb{F}$-submartingales. Every integrable submartingale can be uniquely decomposed into a predictable, non-decreasing process (starting at 0) and a martingale (starting at 0).

4.3 Part 3: The Squared Martingale

Let $S = X^2$ where X is a square integrable $\mathbb{F}$-martingale.

Question 3(a): Show that S is an $\mathbb{F}$-submartingale.

Step-by-Step Explanation: We use Jensen's inequality with the convex function $f(x) = x^2$:

$$E[X_n^2|\mathcal{F}_{n-1}] \geq (E[X_n|\mathcal{F}_{n-1}])^2$$

Because X is a martingale, $E[X_n|\mathcal{F}_{n-1}] = X_{n-1}$. Substituting this in:

$$E[X_n^2|\mathcal{F}_{n-1}] \geq X_{n-1}^2$$

Since $S_n = X_n^2$, this translates to $E[S_n|\mathcal{F}_{n-1}] \geq S_{n-1}$. Thus, S is a submartingale.

Question 3(b): Write $S_n = S_0 + A_n + M_n$ as shown in Doob's decomposition. Show $A_n - A_{n-1} = E[(X_n - X_{n-1})^2|\mathcal{F}_{n-1}]$.

Step-by-Step Explanation: Let's expand the expression we are trying to match:

$$E[(X_n - X_{n-1})^2|\mathcal{F}_{n-1}] = E[X_n^2 - 2X_nX_{n-1} + X_{n-1}^2|\mathcal{F}_{n-1}]$$

Using linearity and the fact that X_{n-1} is known at time $n-1$:

$$= E[X_n^2|\mathcal{F}_{n-1}] - 2X_{n-1}E[X_n|\mathcal{F}_{n-1}] + X_{n-1}^2$$

Since X is a martingale, $E[X_n|\mathcal{F}_{n-1}] = X_{n-1}$. Substitute this in:

$$\begin{aligned} &= E[X_n^2|\mathcal{F}_{n-1}] - 2X_{n-1}(X_{n-1}) + X_{n-1}^2 \\ &= E[X_n^2|\mathcal{F}_{n-1}] - 2X_{n-1}^2 + X_{n-1}^2 \\ &= E[X_n^2|\mathcal{F}_{n-1}] - X_{n-1}^2 \end{aligned}$$

Substituting S back into the equation:

$$= E[S_n|\mathcal{F}_{n-1}] - S_{n-1}$$

We already know from Question 2(a) that $E[S_n|\mathcal{F}_{n-1}] - S_{n-1}$ is exactly the formula for $A_n - A_{n-1}$ in Doob's decomposition. This completes the proof.

5 Exercise 5: Transformation of Martingales

Context: We are given an $\mathbb{F}$-martingale $M = (M_n)_{n\in\mathbb{N}}$ and a bounded $\mathbb{F}$-predictable process $\xi = (\xi_n)_{n\in\mathbb{N}^*}$. We define a new process X as:

$$\begin{aligned} X_0 &:= M_0, \\ X_n &:= M_0 + \sum_{k=1}^{n} \xi_k(M_k - M_{k-1}) \end{aligned}$$

This is the discrete-time equivalent of a stochastic integral, where ξ represents a trading strategy & M represents the price process.

5.1 Part 1: Martingale Property of the Transform

Question 1: Show that $X = (X_n)_{n\in\mathbb{N}}$ is an $\mathbb{F}$-martingale.

Step-by-Step Explanation: To show a process is a martingale, we must verify three properties: integrability, adaptedness, and the martingale conditional expectation property.

1. **Integrability:** X_n is a finite sum of products. Because ξ is a bounded process, there is some constant C such that $|\xi_k| \leq C$. Since M is a martingale, each M_k is integrable. The product of a bounded random variable and an integrable random variable is integrable. Thus, X_n is integrable.

2. **Adaptedness:** For each $k \leq n$, M_k and M_{k-1} are $\mathcal{F}_k$-measurable. Since ξ is predictable, ξ_k is $\mathcal{F}_{k-1}$-measurable (and thus $\mathcal{F}_k$-measurable). Therefore, the entire sum defining X_n is $\mathcal{F}_n$-measurable, making X an adapted process.

3. **Martingale Property:** We need to show that $E[X_n|\mathcal{F}_{n-1}] = X_{n-1}$. It is often easier to look at the increment $X_n - X_{n-1}$ and show its conditional expectation is 0.

   Notice that $X_n - X_{n-1}$ is simply the very last term of the sum:

   $$X_n - X_{n-1} = \xi_n(M_n - M_{n-1})$$

   Taking the conditional expectation:

   $$E[X_n - X_{n-1}|\mathcal{F}_{n-1}] = E[\xi_n(M_n - M_{n-1})|\mathcal{F}_{n-1}]$$

   Because ξ is predictable, ξ_n is known at time $n-1$, so we can pull it out of the conditional expectation:

   $$= \xi_n E[M_n - M_{n-1}|\mathcal{F}_{n-1}]$$

   Because M is a martingale, we know $E[M_n|\mathcal{F}_{n-1}] = M_{n-1}$, which means the expected increment $E[M_n - M_{n-1}|\mathcal{F}_{n-1}]$ is exactly 0.

   $$= \xi_n \times 0 = 0$$

   Therefore, $E[X_n|\mathcal{F}_{n-1}] = X_{n-1}$, proving X is a martingale. Furthermore, taking the unconditional expectation gives $E[X_n] = E[X_0] = E[M_0]$.

5.2 Part 2: A Characterization of Martingales

Question 2: Let $Y = (Y_n)_{n\in\mathbb{N}}$ be $\mathbb{F}$-adapted and integrable. Show that the following two claims are equivalent: (i) Y is an $\mathbb{F}$-martingale; (ii) For any bounded $\mathbb{F}$-predictable process $\theta = (\theta_n)_{n\in\mathbb{N}^*}$, one has $E\left[\sum_{k=1}^{n} \theta_k(Y_k - Y_{k-1})\right] = 0, \forall n \in \mathbb{N}$.

Step-by-Step Explanation: We need to prove implication in both directions.

Proof that (i) implies (ii): Suppose (i) holds, meaning Y is a martingale. Let's look at the expression in (ii). From Part 1, we already proved that if we transform a martingale Y with a bounded predictable process θ, the resulting process is also a martingale. Let $Z_n = \sum_{k=1}^{n} \theta_k(Y_k - Y_{k-1})$. As shown in Part 1, Z is a martingale starting at $Z_0 = 0$. The expected value of any martingale is constant over time, so $E[Z_n] = E[Z_0] = 0$. This exactly matches the formula in (ii). Thus, (i) $\implies$ (ii).

Proof that (ii) implies (i): Suppose (ii) holds for *any* bounded predictable process θ. We want to prove Y is a martingale, which means proving $E[Y_{n+1} - Y_n|\mathcal{F}_n] = 0$. The definition of conditional expectation $E[Y_{n+1} - Y_n|\mathcal{F}_n] = 0$ means that for any arbitrary event $A \in \mathcal{F}_n$, we must have:

$$E[1_A(Y_{n+1} - Y_n)] = 0$$

Because (ii) is true for *any* predictable process, we can custom-build a process θ to isolate exactly this term. Fix an arbitrary time n and an arbitrary set $A \in \mathcal{F}_n$. We define our process θ as follows:

- Let $\theta_{n+1} = 1_A$

- Let $\theta_k = 0$ for all $k \neq n+1$

Is this θ predictable? Yes, because $\theta_{n+1} = 1_A$ depends only on information in $\mathcal{F}_n$. It is also bounded (since indicator functions only take values 0 or 1).

Now plug this custom θ into the equation from (ii) evaluated at time $n+1$:

$$E\left[\sum_{k=1}^{n+1} \theta_k (Y_k - Y_{k-1})\right] = 0$$

Because all θ_k are 0 except for $k = n+1$, the sum collapses into a single term:

$$E[\theta_{n+1}(Y_{n+1} - Y_n)] = 0$$

Substitute $\theta_{n+1} = 1_A$:

$$E[1_A(Y_{n+1} - Y_n)] = 0$$

Since this holds for any $A \in \mathcal{F}_n$, it mathematically forces the conditional expectation $E[Y_{n+1} - Y_n|\mathcal{F}_n]$ to be exactly 0 almost surely. Therefore, $E[Y_{n+1}|\mathcal{F}_n] = Y_n$, proving Y is a martingale.

6 Exercise 6: Change of Measures

Context: Denote by $\mathcal{Q}$ the set of probability measures on $(\Omega, \mathcal{F})$ that are equivalent to P. Two measures are equivalent (written $Q \sim P$) if they agree on what is impossible: $P[A] = 0 \iff Q[A] = 0$.

6.1 Part 1: Convexity of the Set of Equivalent Measures

Question 1: Show that $\mathcal{Q}$ is a convex set.

Step-by-Step Explanation: To show a set is convex, we must take any two elements in the set, say $Q_1, Q_2 \in \mathcal{Q}$, and any weight $\lambda \in [0,1]$, and show that their weighted average (convex combination) $R = \lambda Q_1 + (1-\lambda)Q_2$ is also in the set $\mathcal{Q}$.

First, we check if R is a valid probability measure: 1. $R(\Omega) = \lambda Q_1(\Omega) + (1-\lambda)Q_2(\Omega) = \lambda(1) + (1-\lambda)(1) = 1$. 2. For any event A, since $Q_1(A) \geq 0$ and $Q_2(A) \geq 0$, it follows that $R(A) \geq 0$. 3. Countable additivity holds because it holds for Q_1 and Q_2, and the sum operator is linear. Thus, R is a probability measure.

Next, we check if R is equivalent to P (i.e., $R \sim P$). If $\lambda = 0$, $R = Q_2$, which is equivalent to P. If $\lambda = 1$, $R = Q_1$, which is also equivalent to P. Suppose $\lambda \in (0,1)$. Let $A \in \mathcal{F}$ be an arbitrary event. - If $P[A] = 0$: Because $Q_1 \sim P$ and $Q_2 \sim P$, we must have $Q_1[A] = 0$ and $Q_2[A] = 0$. Therefore, $R[A] = \lambda(0) + (1-\lambda)(0) = 0$. - If $R[A] = 0$: Since $\lambda > 0$, $1 - \lambda > 0$, and probabilities are non-negative, the only way a weighted sum of two non-negative numbers can be zero is if both numbers are zero. Thus, $Q_1[A] = 0$ and $Q_2[A] = 0$. Because $P \sim Q_1$, this implies $P[A] = 0$.

Since $P[A] = 0 \iff R[A] = 0$, the measure R is equivalent to P, so $R \in \mathcal{Q}$. The set $\mathcal{Q}$ is convex.

6.2 Part 2: Making a Variable Integrable

Context: Let X be a positive random variable such that $E[X] = +\infty$. We partition the sample space into sets based on the value of X:

$$A_n := \{\omega \in \Omega : X(\omega) \in [n-1, n)\}, \quad \forall n \in \mathbb{N}^*$$

For $\alpha > 0$, define a random variable Z by:

$$Z(\omega) := \frac{\alpha}{n^3(1 + P[A_n])} \quad \text{if } \omega \in A_n$$

Question 2(a): Choose α such that we may define a probability $Q \in \mathcal{Q}$ satisfying $\frac{dQ}{dP} = Z$.

Step-by-Step Explanation: For Z to be the Radon-Nikodym derivative of a new probability measure Q, we need Q to be a valid probability measure. This requires $Q[\Omega] = 1$, which means the expected value of Z under P must be 1 ($E^P[Z] = 1$).

Let's compute $E^P[Z]$:

$$E^P[Z] = \sum_{n=1}^{\infty} E^P[Z 1_{A_n}]$$

Because Z is constant on each set A_n, we can pull it out:

$$= \sum_{n=1}^{\infty} \frac{\alpha}{n^3(1+P[A_n])} E^P[1_{A_n}] = \sum_{n=1}^{\infty} \frac{\alpha P[A_n]}{n^3(1+P[A_n])}$$

Let $C = \sum_{n=1}^{\infty} \frac{P[A_n]}{n^3(1+P[A_n])}$. Notice that the fraction $\frac{P[A_n]}{1+P[A_n]}$ is always less than 1. Therefore, the terms in the sum are strictly less than $\frac{1}{n^3}$. Because the series $\sum \frac{1}{n^3}$ is convergent (p-series with $p = 3 > 1$), our sum C is finite ($C < \infty$). To make $E^P[Z] = 1$, we simply choose $\alpha = C^{-1}$. Since $P \geq 0$, $\alpha > 0$. Because $Z > 0$ almost everywhere, the measure Q is equivalent to P ($Q \sim P$).

Question 2(b): Show that $X \in L^1(\Omega, \mathcal{F}, Q)$.

Step-by-Step Explanation: We want to prove $E^Q[X] < \infty$. Using the change of measure formula, $E^Q[X] = E^P[ZX]$.

$$E^P[ZX] = \sum_{n=1}^{\infty} E^P[ZX 1_{A_n}]$$

On the set A_n, we know two things: $Z = \frac{\alpha}{n^3(1+P[A_n])}$ and $X < n$. Substituting the exact value for Z and the upper bound for X:

$$E^P[ZX 1_{A_n}] \leq E^P\left[\frac{\alpha}{n^3(1+P[A_n])} n 1_{A_n}\right] = \frac{\alpha n}{n^3(1+P[A_n])} P[A_n] = \frac{\alpha P[A_n]}{n^2(1+P[A_n])}$$

Again, noting that $\frac{P[A_n]}{1+P[A_n]} < 1$, we get:

$$E^P[ZX 1_{A_n}] \leq \frac{\alpha}{n^2}$$

Summing this over all n:

$$E^Q[X] \leq \sum_{n=1}^{\infty} \frac{\alpha}{n^2}$$

Because $\sum \frac{1}{n^2}$ is a convergent p-series ($p = 2 > 1$), the total expectation under Q is finite. Thus, $X \in L^1(\Omega, \mathcal{F}, Q)$.

Question 2(c): Summarize the result. For every positive random variable X on $(\Omega, \mathcal{F}, P)$, even if its expectation is infinite, there exists an equivalent probability measure Q under which X has a finite expectation (it is integrable). If X is already integrable under P, we can just choose $Q = P$.

6.3 Part 3: Preservation of Strict Positivity

Question 3: Let $Q \in \mathcal{Q}$ and $X \in L^1(\Omega, \mathcal{F}, P)$. Assume $X \geq 0$ P-a.s. and $P[X > 0] > 0$. Show that these properties are still satisfied for Q.

Step-by-Step Explanation: 1. **Almost sure non-negativity:** The statement "$X \geq 0$ P-a.s." means the probability of the opposite event is zero: $P[\{\omega \in \Omega : X(\omega) < 0\}] = 0$. Because Q and P are equivalent, they agree on probability-zero events. Therefore, $Q[\{\omega \in \Omega : X(\omega) < 0\}] = 0$, which means $X \geq 0$ Q-a.s. 2. **Strict positivity of probability:** We are given $P[X > 0] > 0$. We want to prove $Q[X > 0] > 0$. We can use a proof by contradiction. Suppose $Q[X > 0] = 0$. By the equivalence of the two measures, any event with Q-probability 0 must also have P-probability 0. This would force $P[X > 0] = 0$, which contradicts our initial assumption. Thus, it must be that $Q[X > 0] > 0$.

7 Exercise 7: Radon-Nikodym's Derivative

Context: Let Q be a probability measure on $(\Omega, \mathcal{F})$ that is equivalent to P and let $\mathcal{G} \subset \mathcal{F}$ be a sub-algebra.

7.1 Part 1: Equivalence on the Sub-algebra

Question 1: Show that P & Q are equivalent on $(\Omega, \mathcal{G})$.

Step-by-Step Explanation: To show that P & Q are equivalent on the restricted space $(\Omega, \mathcal{G})$, we need to prove that for any event $A \in \mathcal{G}$, $P[A] = 0$ if and only if $Q[A] = 0$.

Because $\mathcal{G}$ is a sub-algebra of $\mathcal{F}$, any event A that belongs to $\mathcal{G}$ automatically belongs to $\mathcal{F}$ ($A \in \mathcal{G} \implies A \in \mathcal{F}$). We already know that P & Q are equivalent on the larger space $(\Omega, \mathcal{F})$. This means that for any event in $\mathcal{F}$, they agree on what has a probability of zero. Since every event in $\mathcal{G}$ is also in $\mathcal{F}$, the equivalence naturally carries over. Therefore, P & Q are equivalent on $(\Omega, \mathcal{G})$.

Because of this equivalence, we can properly denote the Radon-Nikodym derivative defined specifically on $(\Omega, \mathcal{G})$ as $\frac{dQ}{dP}\big|_{\mathcal{G}}$.

7.2 Part 2: Conditional Expectation Connection

Question 2: Show that $\frac{dQ}{dP}\big|_{\mathcal{G}} = E^P\left[\frac{dQ}{dP}\middle|\mathcal{G}\right]$ P-a.s.

Step-by-Step Explanation: To make the notation cleaner, let's define two variables: Let $A = \frac{dQ}{dP}\big|_{\mathcal{G}}$ (the derivative restricted to $\mathcal{G}$). Let $B = \frac{dQ}{dP}$ (the full derivative on $\mathcal{F}$).

We want to prove that $A = E^P[B|\mathcal{G}]$. By the mathematical definition of conditional expectation, $E^P[B|\mathcal{G}]$ is the unique $\mathcal{G}$-measurable random variable that satisfies the following property for every bounded $\mathcal{G}$-measurable test variable Z:

$$E^P[BZ] = E^P\left[E^P[B|\mathcal{G}]Z\right]$$

Let's compute the expected value of Z under the measure Q in two different ways, using our two different Radon-Nikodym derivatives.

Path 1 (Using B): Because Z is $\mathcal{G}$-measurable, it is also $\mathcal{F}$-measurable. Using the full derivative B to change the measure from P to Q over $\mathcal{F}$, we get:

$$E^Q[Z] = E^P[BZ]$$

Path 2 (Using A): Because Z is $\mathcal{G}$-measurable, we can use the restricted derivative A to change the measure from P to Q directly over $\mathcal{G}$:

$$E^Q[Z] = E^P[AZ]$$

Conclusion: Since both paths compute the exact same expected value $E^Q[Z]$, we can equate them:

$$E^P[AZ] = E^P[BZ]$$

By definition, A is $\mathcal{G}$-measurable because it is the Radon-Nikodym derivative defined on the sub-algebra $\mathcal{G}$. Since A is $\mathcal{G}$-measurable and satisfies $E^P[AZ] = E^P[BZ]$ for every bounded $\mathcal{G}$-measurable random variable Z, A perfectly satisfies the definition of the conditional expectation of B given $\mathcal{G}$.

Due to the uniqueness of conditional expectation (up to a P-null set), we conclude:

$$A = E^P[B|\mathcal{G}] \quad P\text{-a.s.}$$

Which translates exactly to our target equation:

$$\left.\frac{dQ}{dP}\right|_{\mathcal{G}} = E^P\left[\frac{dQ}{dP}\middle|\mathcal{G}\right] \quad P\text{-a.s.}$$